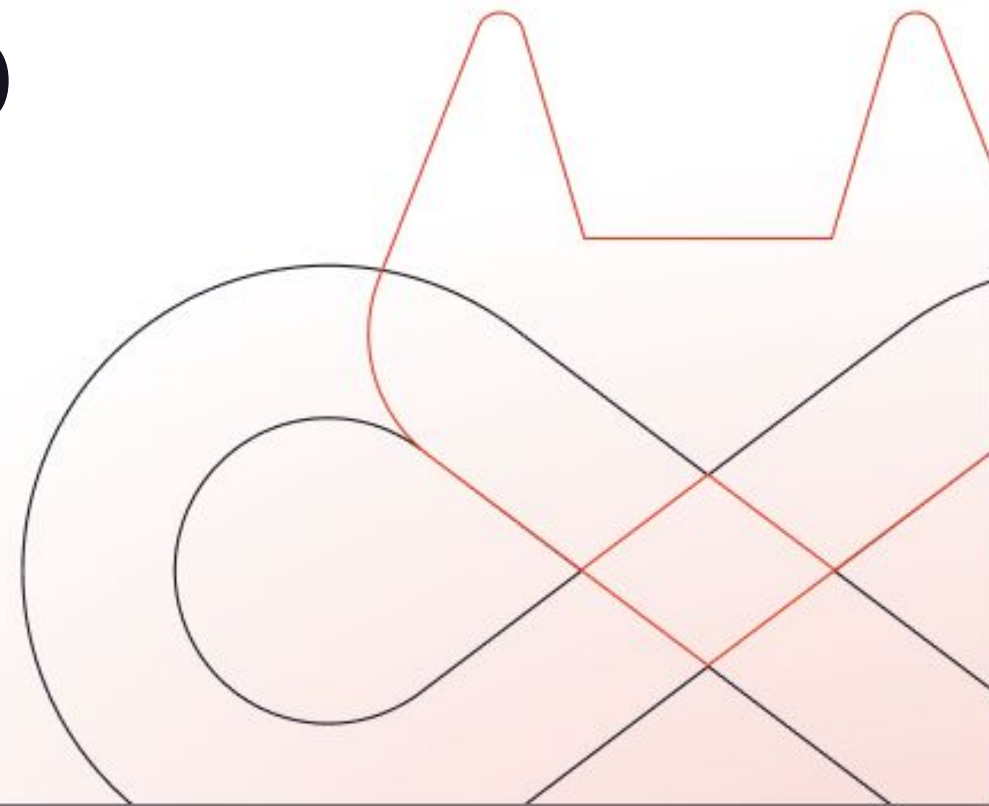




# Advanced GitLab CI/CD





1. **CI/CD Basics: a quick review**
2. **Pipeline structure and ways to hack it**  
Pipeline Architectures
3. **Variables: a foundation before using rules**
4. **Controlling when your jobs run**  
The “rules” label
5. **Holding, securing, and sharing results**  
Job Artifacts
6. **Assembling pipelines from components**  
Using “includes” and “extends” labels

This workshop assumes that participants are familiar with basic GitLab CI/CD.

It presents advanced concepts and techniques for people who are active users of GitLab CI.

We do not cover Runner configuration and any of the CD-specific features such as Environments and Deployments.



# GitLab CI/CD Basics Review



# GitLab Recommended Process



Manage



Plan



Create



Verify



Package



Secure



Release



Configure



Monitor



Protect



Epics



Milestones



Issues

Create Merge Request



Push Code



Automated Build / Test



Scan

Collaboration & Review



Review App



Approval



Push Fixes



Merge Accepted

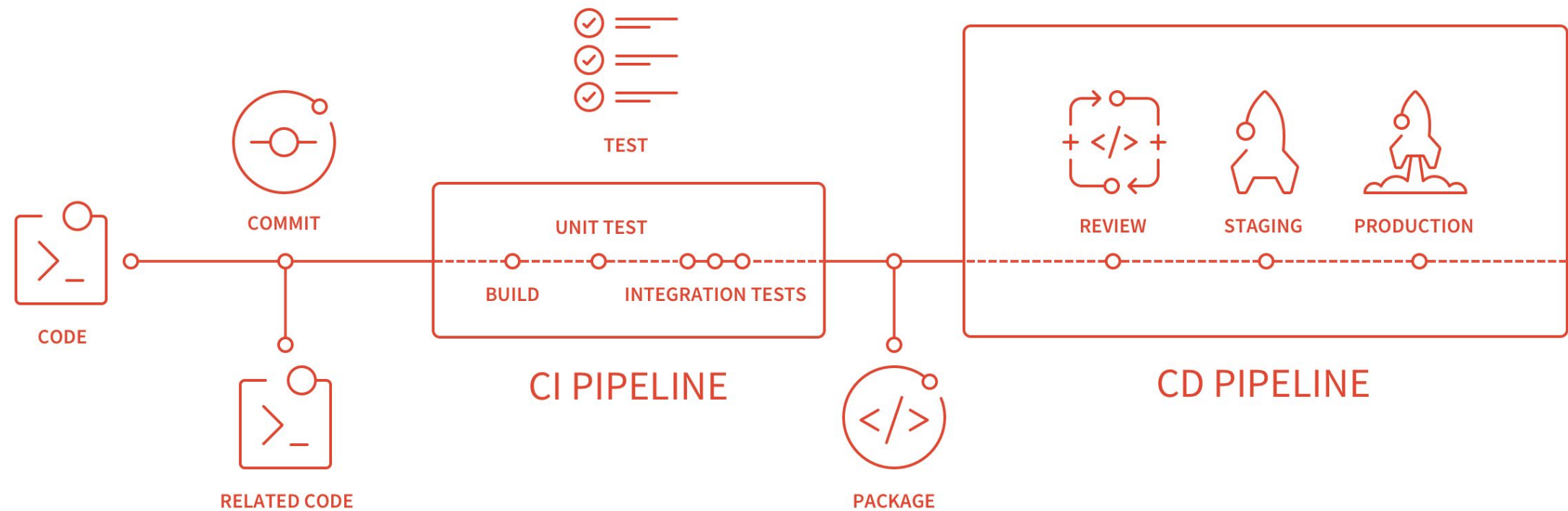


Release



Deploy







## Pipeline

- Set of one or more jobs. Optionally organized into stages

## Stages

- Collection of jobs to be run in parallel
- e.g. Build, Test, Deploy

## Jobs

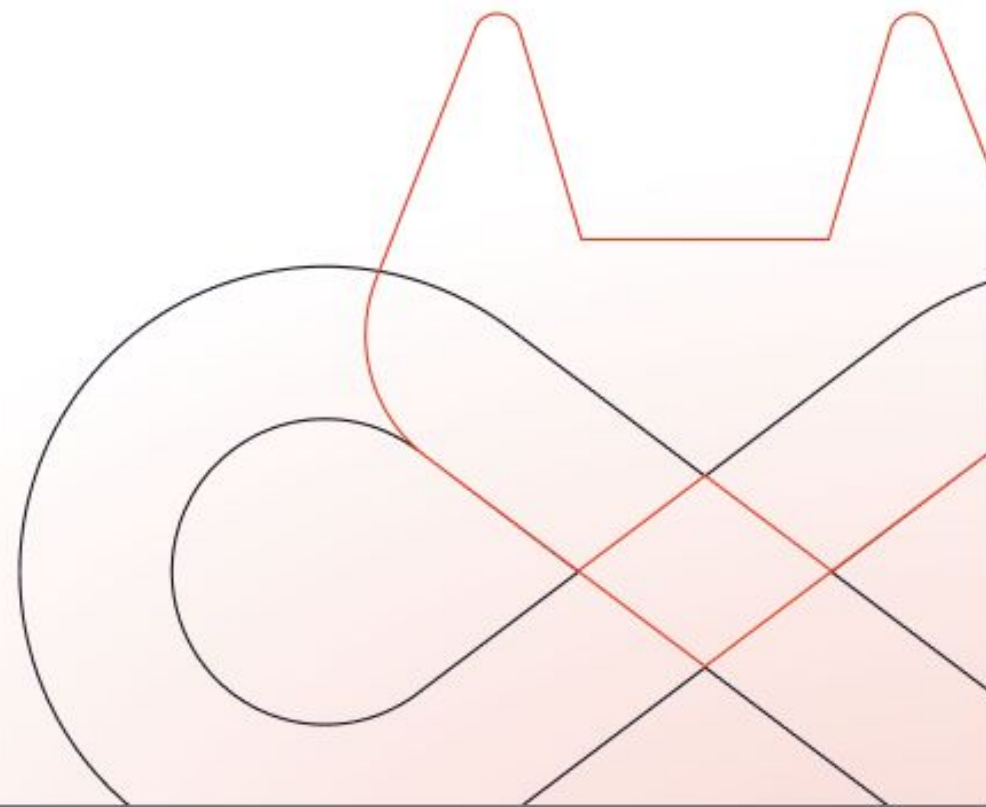
- Scripts that perform tasks
- e.g. npm test; mvn install; etc.

## Environments

- Where we deploy (Test, Review, Staging, Canary, Prod)



# Pipeline Architectures





- The building-blocks of your pipeline.
  - [Basic](#)
  - [Directed Acyclic Graph](#)
  - [Parent/Child Pipelines](#)
  - [Dynamic Child Pipelines](#)
  - [Multi-Project Pipelines](#)



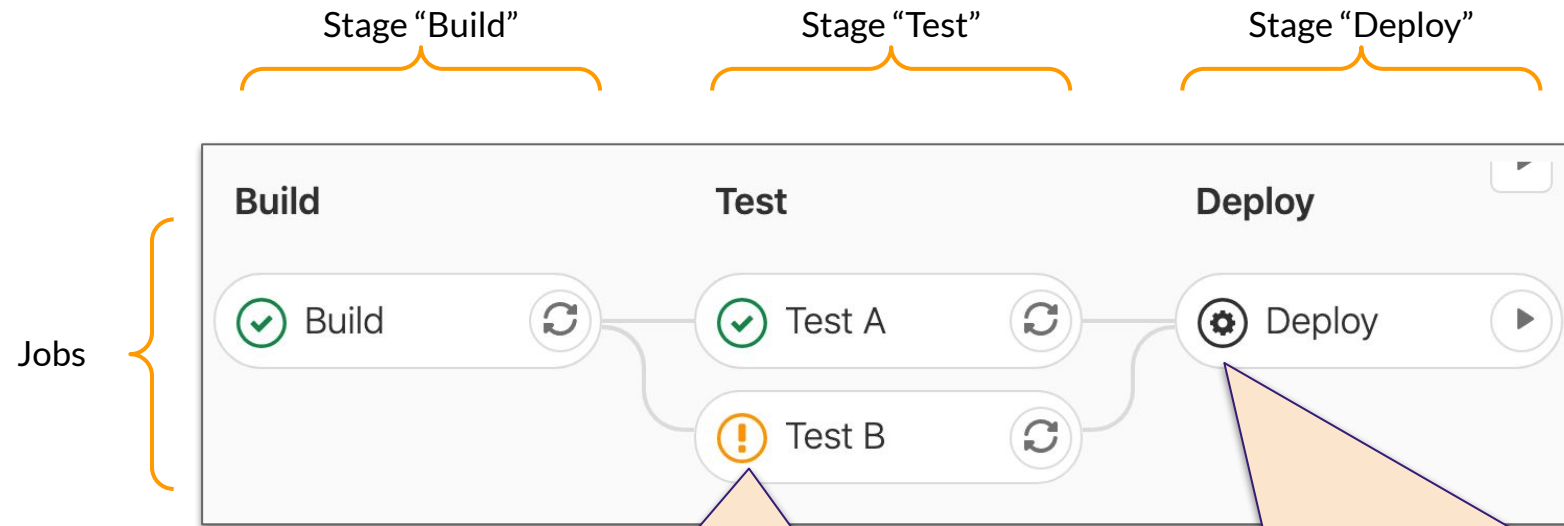




# Basic pipelines

- Simplest pipeline in GitLab
- Jobs run independently, sometimes on different runners
- All Jobs in a Stage must complete successfully before proceeding to the next stage
- Control pipeline with pipeline options

# Pipeline Architectures - Basic pipeline with options



Other options:

**when: delayed**  
**when: on\_failure**  
**when: always**

This job has

**allow\_failure: true**

...so the pipeline proceeds even though it failed

This job has

**when: manual**

...so it waits for someone (with the right permissions) to click a "Play" button





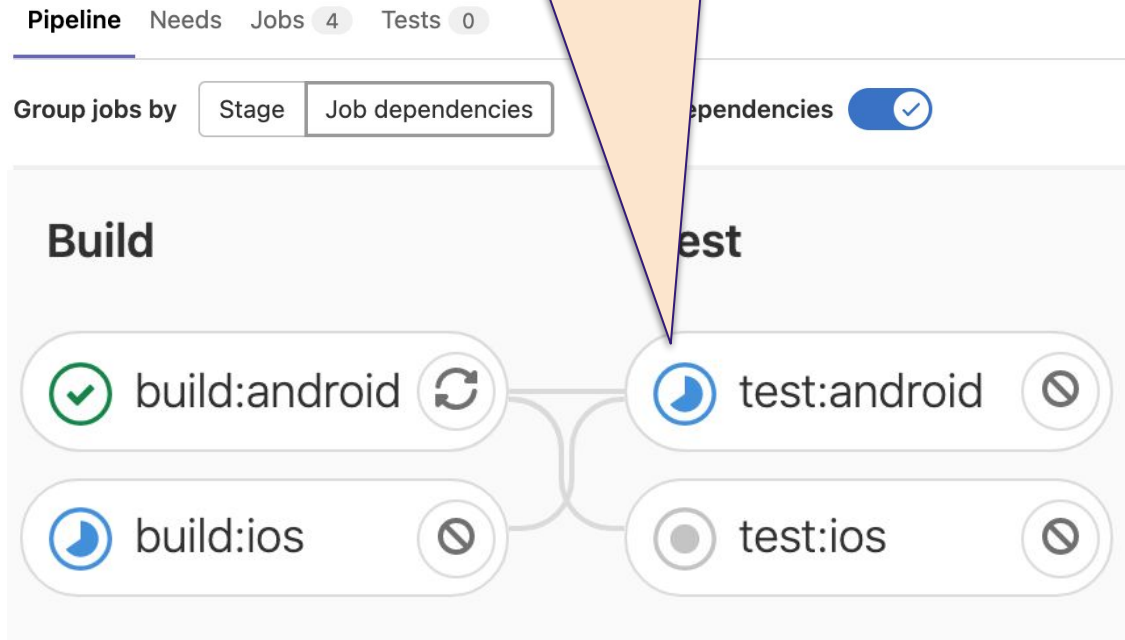
# Needs or Directed Acyclic Graph (DAG)

- Define job dependencies to optimize pipeline flow
- Jobs still run independently, sometimes on different runners
- Dependent jobs can proceed to next stage without waiting for other jobs in stage to finish

# Pipeline Architectures - Directed Acyclic Graph (DAG)



The Android test doesn't have to wait for the iOS build to complete



```
build:ios:
  stage: build
  script:
    - echo "Building for iOS"
    - sleep 10
```

```
build:android:
  stage: build
  script:
    - echo "Building for Android"
```

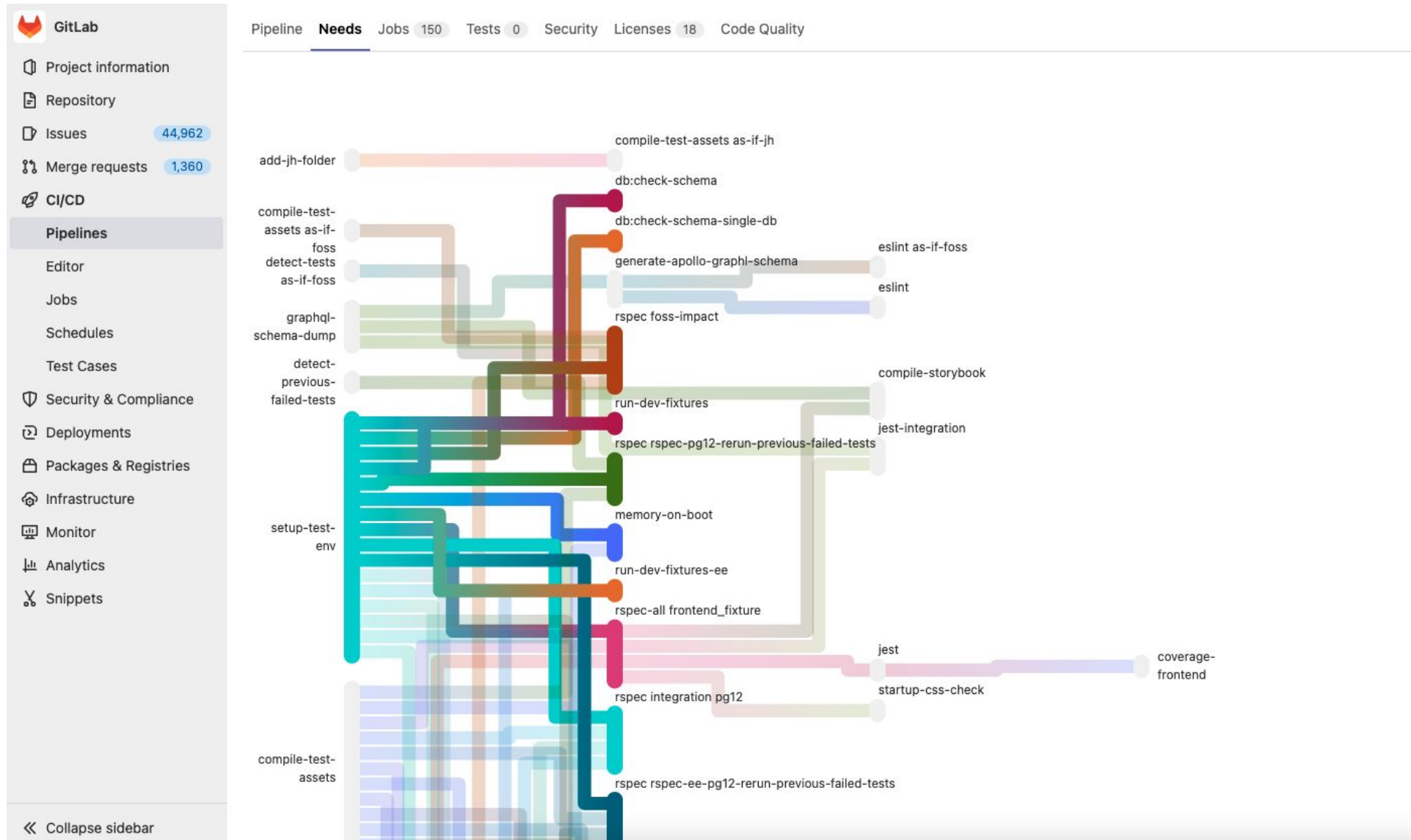
```
test:ios:
  stage: test
  needs: ["build:ios"]
  script:
    - echo "Testing for iOS"
```

```
test:android:
  stage: test
  needs: ["build:android"]
  script:
    - echo "Testing for Android"
```

Jobs still runs in stages, but the "needs" keyword overrides the need for the previous stage to complete entirely.



# Directed Acyclic Graph Pipelines



# Example: needs



```
linux:build:
  stage: build
  script: echo "Building linux..."

mac:build:
  stage: build
  script: echo "Building mac..."

lint:
  stage: test
  needs: []
  script: echo "Linting..."

linux:rspec:
  stage: test
  needs: ["linux:build"]
  script: echo "Running rspec on linux..."

mac:rspec:
  stage: test
  needs: ["mac:build"]
  script: echo "Running rspec on mac..."

production:
  stage: deploy
  script: echo "Running production..."
```

Linux path: the **linux:rspec** job will be run as soon as the **linux:build** job finishes without waiting for **mac:build** to finish.

macOS path: the **mac:rspec** job will be run as soon as the **mac:build** job finishes without waiting for **linux:build** to finish.

The **lint** job will run immediately without waiting for **build** jobs.

The **production** job runs as soon as all previous jobs finish.

Source: <https://docs.gitlab.com/ee/ci/yaml/#needs>

# Additional example: needs with artifacts



```
test-job1:
  stage: test
  needs:
    - job: build_job1
      artifacts: true

test-job2:
  stage: test
  needs:
    - job: build_job2
      artifacts: false

test-job3:
  needs:
    - job: build_job1
      artifacts: true
    - job: build_job2
    - build_job3
```

The `test-job1` job downloads the `build_job1` artifacts

The `test-job2` job does not download the `build_job2` artifacts.

The `test-job3` job downloads the artifacts from all three `build_jobs`, because `artifacts` is `true`, or defaults to `true`, for all three needed jobs.

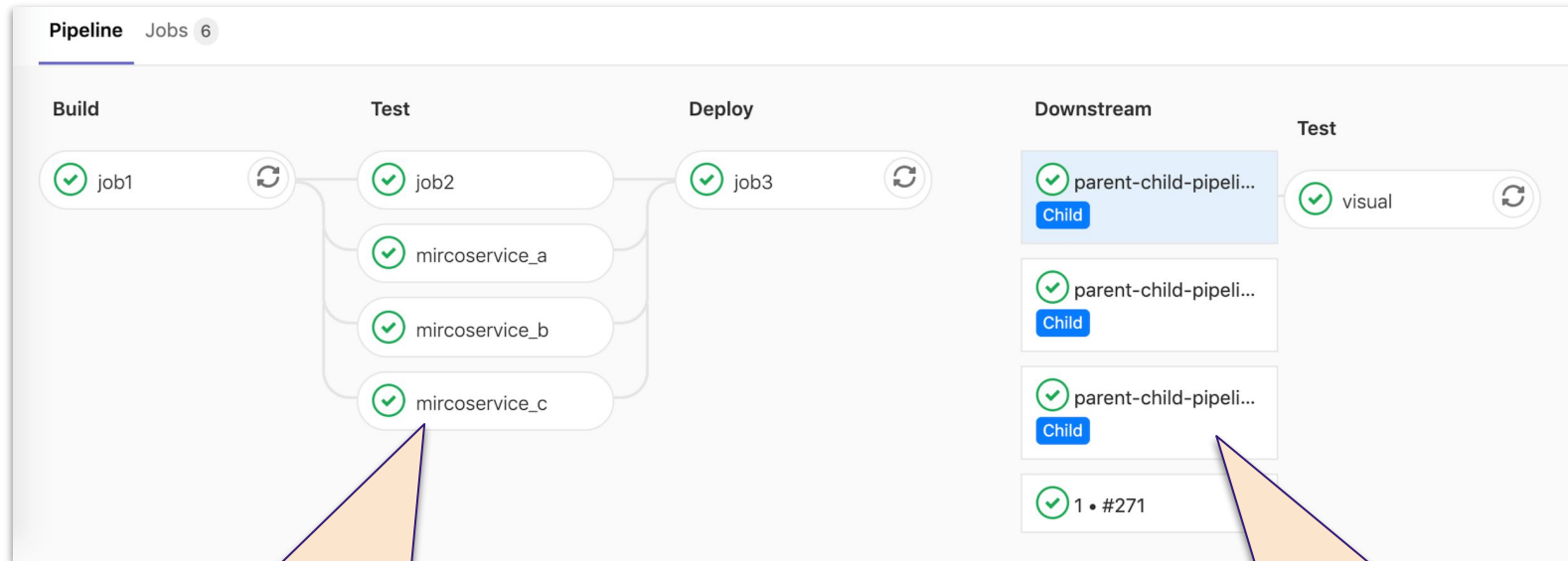


# Parent/Child Pipelines

- Run child pipelines independent from each other.
- Separates entire pipeline configuration in multiple files to keep things simple.
- Combine with DAG pipelines to achieve benefits of both.
- Useful to branch out long running tasks into separate pipelines.



# Pipeline Architectures - Parent/Child Pipelines



The pipeline you're viewing

Downstream pipelines that are triggered from within this one.



# Pipeline Architectures - Parent/Child Pipelines for Monorepos



Run the job if there are changes in those files

```
stages:
  - build
  - finalize

build-1:
  stage: build
  only:
    changes:
      - project-1/*
  trigger:
    include: project-1/.gitlab-ci.yml
    strategy: depend
```

Include files from elsewhere in the project - must reference a YAML file

**strategy: depend**  
means to hold this pipeline until the other pipeline finishes





# Dynamic Pipelines

- Generate pipeline configuration at build time
- Use the generated configuration at a later stage to run as a child pipeline.
- Useful to use a single pipeline configuration with different settings to support a matrix of targets and architectures

# Pipeline Architectures - Dynamic Pipelines



## .gitlab-ci.yml

```
stages:
  - setup
  - test
  - finalize

setup:
  stage: setup
  script:
    - sed "s/REPLACEME/DynamicPipeline/g" test.gitlab-ci.yml > generated-config.yml
  artifacts:
    paths:
      - generated-config.yml

test:
  stage: test
  trigger:
    include:
      - artifact: generated-config.yml
        job: setup
    strategy: depend

finalize:
  stage: finalize
  script:
    - echo "Finalize job"
```

Place a generated YAML file in the job artifact store

Reference it later to actually run the pipeline

## test.gitlab-ci.yml

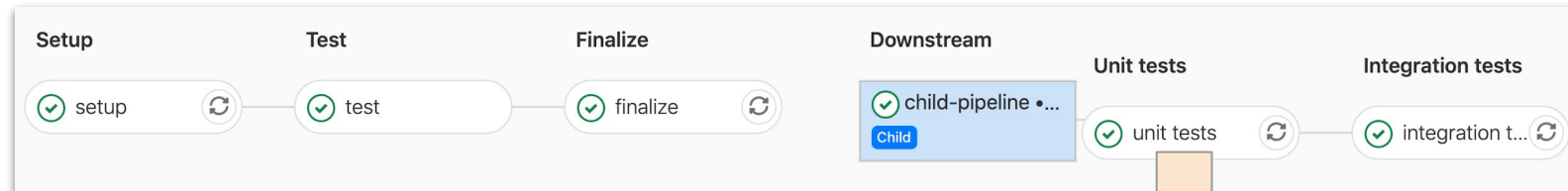
```
stages:
  - unit tests
  - integration tests

unit tests:
  stage: unit tests
  script:
    - echo "REPLACEME"

integration tests:
  stage: integration tests
  script:
    - echo "REPLACEME"
```



# Pipeline Architectures - Dynamic Pipelines



passed Job #673072268 triggered 9 minutes ago by Francis Potter

```
1 Running with gitlab-runner 13.2.2 (a998cacd)
2 on docker-auto-scale fa6cab46
3 Preparing the "docker+machine" executor
4 Using Docker executor with image ruby:2.5 ...
5 Pulling docker image ruby:2.5 ...
6 Using docker image sha256:896afe8ad24a941448575390111af68ddcf195105b8d1f23e60461c0b8c4ed3c for ruby:2.5 ...
7 Preparing environment
8 Running on runner-fa6cab46-project-20364915-concurrent-0 via runner-fa6cab46-srm-1596667784-f3980b85...
9 Getting source from Git repository
10 $ eval "$CI_PRE_CLONE_SCRIPT"
11 Fetching changes with git depth set to 50...
12 Initialized empty Git repository in /builds/fpotter/examples/dynamic-pipelines/.git/
13 Created fresh repository.
14 Checking out 1bf7ef6f as master...
15 Skipping Git submodules setup
16 Executing "step_script" stage of the job script
17 $ echo "DynamicPipeline"
18 DynamicPipeline
19 Job succeeded
```

The replacement happened in the parent pipeline





# Multi-project Pipelines

- A pipeline in one project can trigger a pipeline in another project
- You can specify a specific branch
- You can pass variables to downstream pipelines
- If downstream pipeline fails it will not fail upstream pipeline
- Useful when building / deploying large applications that are made up of different components that have their own project and build pipeline

Documentation link:

[https://docs.gitlab.com/ee/ci/multi\\_project\\_pipelines.html](https://docs.gitlab.com/ee/ci/multi_project_pipelines.html)

# Pipeline Architectures - Multi-Project Pipelines



```
test:
  stage: test
  script:
    - echo "I am building!"

bridge:
  variables:
    ENVIRONMENT: staging
  stage: test
  trigger:
    project: my/project
    branch: master
    strategy: depend
    forward:
      pipeline_variables:
```

Pass variables to downstream projects

Specify project and branch

Define what variables to forward

The job with the trigger is referred to as the "bridge" job

Downstream pipeline can be anything

```
staging:
  stage: deploy
  script:
    - echo "Deploying to $ENVIRONMENT!"
```

Build

Test

Downstream

Deploy

✓ test



✓ bridge

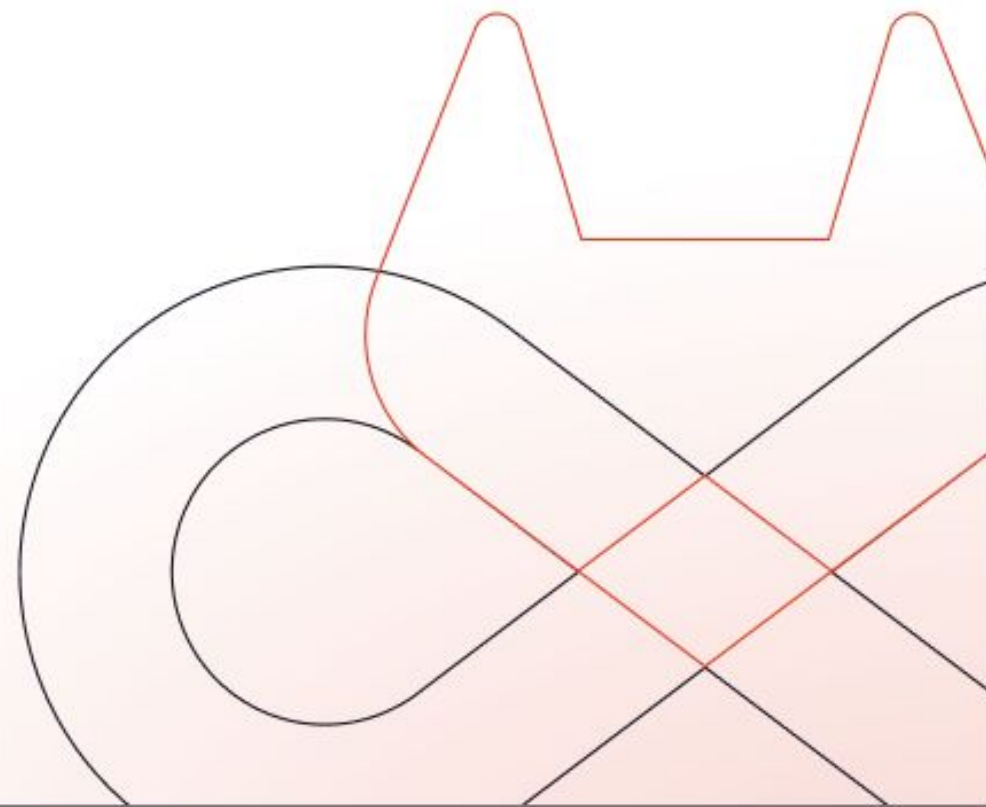
✓ Pipeline - multi...

Multi-project

✓ staging



# Variables





# Variables in UI at project level



chipmunk

Project overview

Repository

Issues 30

Merge Requests 3

Requirements

CI / CD

Security & Compliance

Operations

Packages & Registries

Analytics

Wiki

Snippets

Members

**Settings**

General

Integrations

Webhooks

Repository

**CI / CD**

Operations

Pages

Audit Events

Variables ?

Environment variables are applied to environments via the Runner. You can use environment variables for passwords, secret keys, etc. Make variables available to the running application by prepending the variable key with `K8S_SECRET_`. You can set variables to be:

- Protected variables are only exposed to protected branches or tags.
- Masked variables are hidden in job logs (though they must match certain regexp requirements to do so).

[More information](#)

Environment variables are configured by your administrator to be protected by default

| Type     | Key            | Value | Protected | Masked | Environments                    |
|----------|----------------|-------|-----------|--------|---------------------------------|
| Variable | CI_DEBUG_TRACE | ***** | ✓         | ✗      | All (default) <a href="#">✎</a> |
| Variable | SAST_DISABLED  | ***** | ✓         | ✗      | All (default) <a href="#">✎</a> |

[Reveal values](#) [Add Variable](#)

**Group variables (inherited)**

These variables are configured in the parent group settings, and are inherited by this project in addition to the project variables.

| Key | Origin |
|-----|--------|
|-----|--------|

**Pipeline triggers** [Expand](#)

Triggers can force a specific branch or tag to get rebuilt with an API call. These tokens will impersonate their associated user including their access to projects and their project permissions.

**Cleanup policy for tags** [Expand](#)

Save space and find tags in the Container Registry more easily. Enable the cleanup policy to remove stale

Variables can also be set at the Group and Instance level



# Variables entered for the pipeline run



GitLab Pipelines page for the 'chipmunk' project. The 'CI / CD' menu item is circled in red. The 'Run Pipeline' button is also circled in red. A large orange arrow points from the 'Run Pipeline' button to the 'Run Pipeline' modal window on the right.

| Status  | Pipeline   | Triggerer | Commit                   | Stages               |
|---------|------------|-----------|--------------------------|----------------------|
| skipped | #160244003 | latest    | Ptest-bra...<br>ce1aa56c | Add CODEOWNERS       |
| failed  | #158950442 | latest    | P41-chang...<br>cf5d1d26 | Update db/demo.txt   |
| skipped | #158950354 | latest    | P41-chang...<br>ce1aa56c | Add CODEOWNERS       |
| skipped | #158947065 | latest    | Pmaster -> ce1aa56c      | Add CODEOWNERS       |
| failed  | #153772549 | latest    | Pmaster -> 2eccca75      | Merge branch '8-m... |
| passed  | #153052255 | latest    | P7-est-te...<br>8980e09d | Est tempora sit d... |
| failed  | #153051428 | latest    | P7-est-te...<br>2eccca75 | Merge branch '8-m... |
| passed  | #152990369 | latest    | P10-non-d...<br>2eccca75 | Merge branch '8-m... |

**Run Pipeline**

Run for  
master

Existing branch name or tag

Variables

Variable  Input variable key  Input variable

Specify variable values to be used in this run. The values specified in C

**Run Pipeline**

## Hot tip!

Prepopulate the keys and values using URL parameters

.../pipelines/new?ref=<branch>&var[<variable\_key>]=<value>



# Variables defined in the CI configuration



```
variables:  
  GLOBAL_VAR: test  
  
build:  
  variables:  
    ENVIRONMENT: staging  
  script:  
    - echo "Trying $GLOBAL_VAR on $ENVIRONMENT"
```



# Secrets



```
deploy:
  stage: deploy
  secrets:
    DATABASE_PASSWORD:
      vault: # Translates to secret: `ops/data/production/db`, field: `password`
      engine:
        name: kv-v2
        path: ops
        path: production/db
        field: password
      file: false

# Shortened version:
# vault: production/db/password@ops
```



# Inherited variables

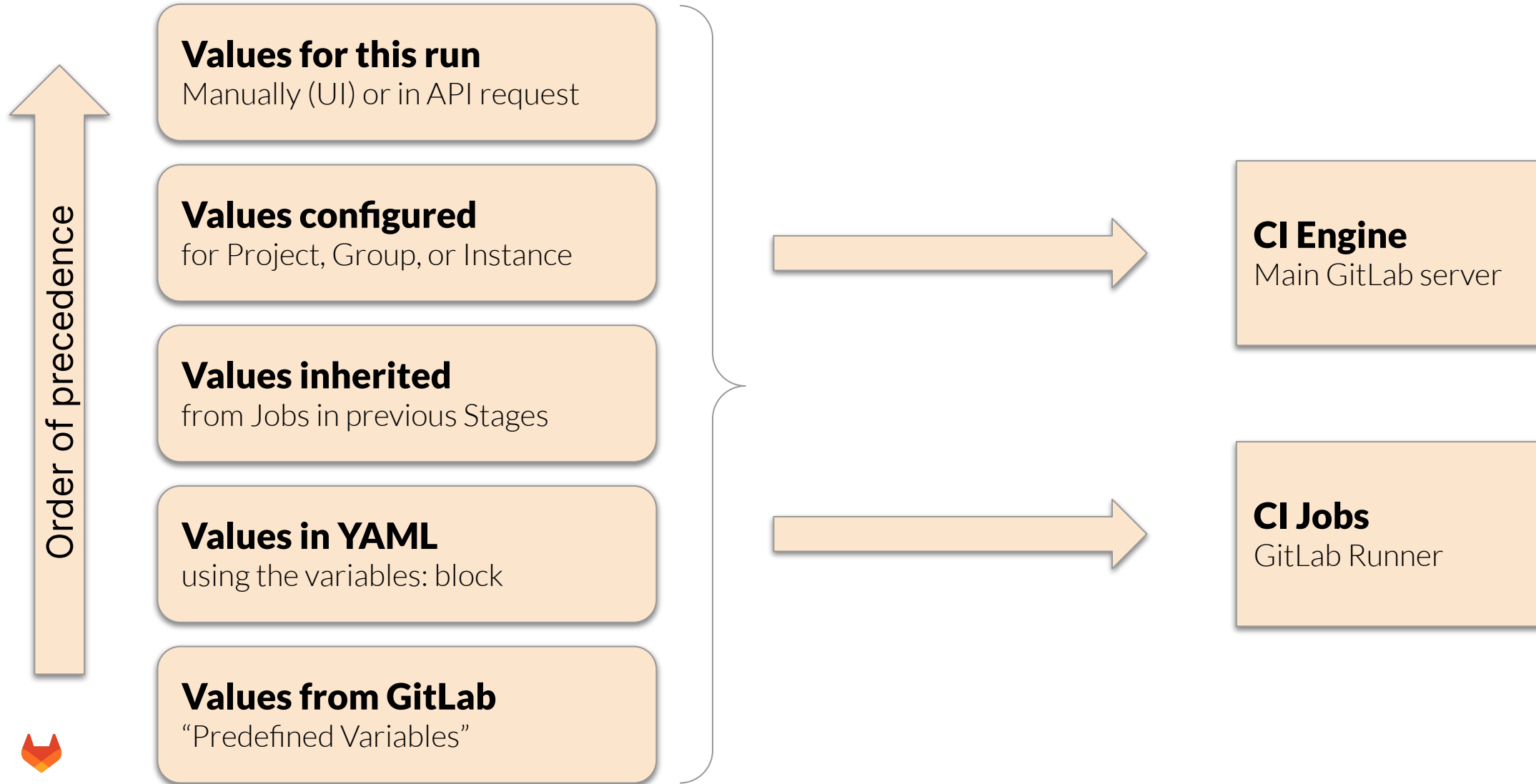


```
build:
  stage: build
  script:
    - echo "BUILD_VARIABLE=value_from_build_job" >> build.env
  artifacts:
  reports:
  dotenv: build.env

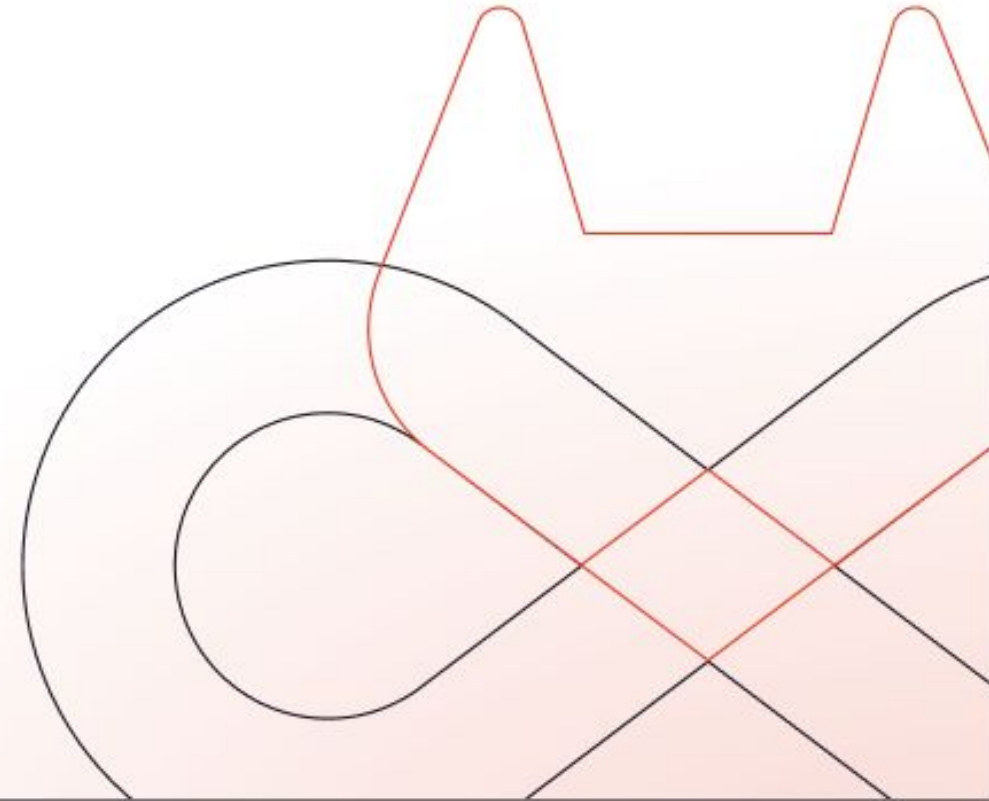
deploy:
  stage: deploy
  variables:
    BUILD_VARIABLE: value_from_deploy_job
  script:
    - echo "$BUILD_VARIABLE" # Output is: 'value_from_build_job' due to precedence
```



# How variables are processed



# Rules



# When are pipelines run?



New commit

New branch

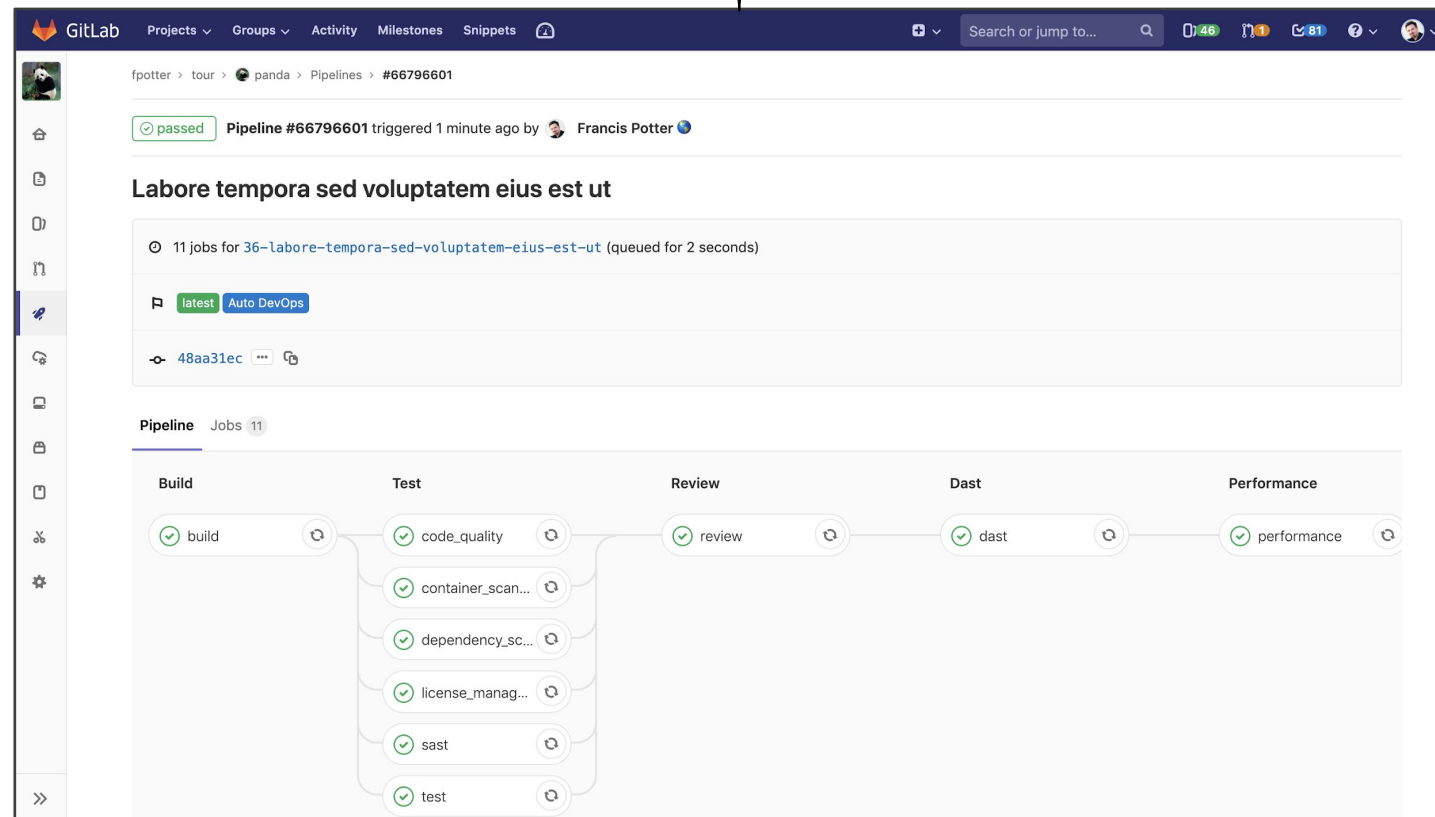
New tag

New merge  
request

Manual

API call

Scheduled





# Rules: The basics



Job Name

New “rules”  
block

```
only-on-web-run:  
  rules:  
    - if: $CI_PIPELINE_SOURCE=="web"  
  script:  
    - echo "This job will only run when the pipeline is triggered from the web form"
```

“If” statements can reference  
Variables - including the  
predefined ones, as in this case

As it claims, this job will only  
run when the pipeline is  
kicked off from the web form



# Rules Quick Reference



## Clauses

if  
changes  
exists  
(... or none)

## Operators

*(just the variable)*

==

!=

=~

!~

&&

||

## Results

when  
allow\_failure  
start\_in

## “when” options

always  
never  
on\_success  
on\_failure  
manual  
Delayed  
(... or none)





## Default values

```
when: on_success  
allow_failure: false
```

## Job is added when:

A rule matches and has either:

- `when: on_success`
- `when: delayed`
- `when: always`

No match, but last clause is:  
**(As *standalone attribute*)**

- `when: on_success`
- `when: delayed`
- `when: always`

## Job is not added when:

No rules matches and no standalone:

- `when: on_success`
- `when: delayed`
- `when: always`

A rule matches with:

- `when: never`



# Rules Example 01



```
job:
  Script: "echo Hello, Rules!"
  rules:
    - if: '$CI_PIPELINE_SOURCE == "merge_request_event"'
    - if: '$CI_PIPELINE_SOURCE == "push"'
```

- Control when merge request pipelines run.
  - `if: '$CI_PIPELINE_SOURCE == "merge_request_event"'`
- Control when both branch pipelines and tag pipelines run.
  - `if: '$CI_PIPELINE_SOURCE == "push"'`



# Rules Example 02



```
job:
  script: "echo Hello, Rules!"
  rules:
    - if: '$CI_PIPELINE_SOURCE == "merge_request_event"'
      when: never
    - if: '$CI_PIPELINE_SOURCE == "schedule"'
      when: never
    - when: on_success
```

- This job will not run if this was triggered from a merge request
  - if: '\$CI\_PIPELINE\_SOURCE == "merge\_request\_event"'
- This job will not run if pipeline was scheduled
  - if: '\$CI\_PIPELINE\_SOURCE == "schedule"'
- Otherwise, this job will run if the previous stage was successful



# Rules Example 03



```
job:
  script: "echo This creates double pipelines!"
  rules:
    - if: '$CUSTOM_VARIABLE == "false"'
      when: never
    - when: always
```

- This job will not run if `CUSTOM_VARIABLE` is equal to false
  - if: '\$CUSTOM\_VARIABLE == "false"'
- Otherwise, this job will run even if the previous stage had a failure



# Rules Example 04



```
docker build:
  script: docker build -t my-image:$CI_COMMIT_REF_SLUG .
  rules:
    - if: '$VAR == "string value"'
      changes: # Will include the job and set to when:manual if any of the follow paths match a modified fi
        - Dockerfile
        - docker/scripts/*
      when: manual
# - when: never would be redundant here, this is implied any time rules are listed.
```

- This job will be in the pipeline and will be a manual job
  - if: '\$VAR == "string value"' and there was a change to Dockerfile or any file in docker/scripts
- Otherwise, this job will not run



# Rules Example 05



```
docker build:  
  script: docker build -t my-image:$CI_COMMIT_REF_SLUG .  
  rules:  
    - if: '$CI_COMMIT_BRANCH == "main"'  
      when: delayed  
      start_in: '3 hours'  
      allow_failure: true
```

- This job will run 3 hours after triggered and will be allowed to fail (will not prevent further stages from firing)
  - if: '\$CI\_COMMIT\_BRANCH == "main"'
- Otherwise, this job will not run





# Workflow Rules



```
workflow:
  rules:
    - if: $CI_COMMIT_MESSAGE =~ /-wip$/
      when: never
    - if: $CI_COMMIT_TAG
      when: never
    - when: always
```

- This pipeline will not run if the commit message ends with “-wip”
  - if: '\$CI\_COMMIT\_MESSAGE =~ /-wip\$/'
- This pipeline will not run if this was triggered by a TAG being applied
  - if: '\$CI\_COMMIT\_TAG'
- Otherwise, this pipeline will run



# Workflow - Variables depending on

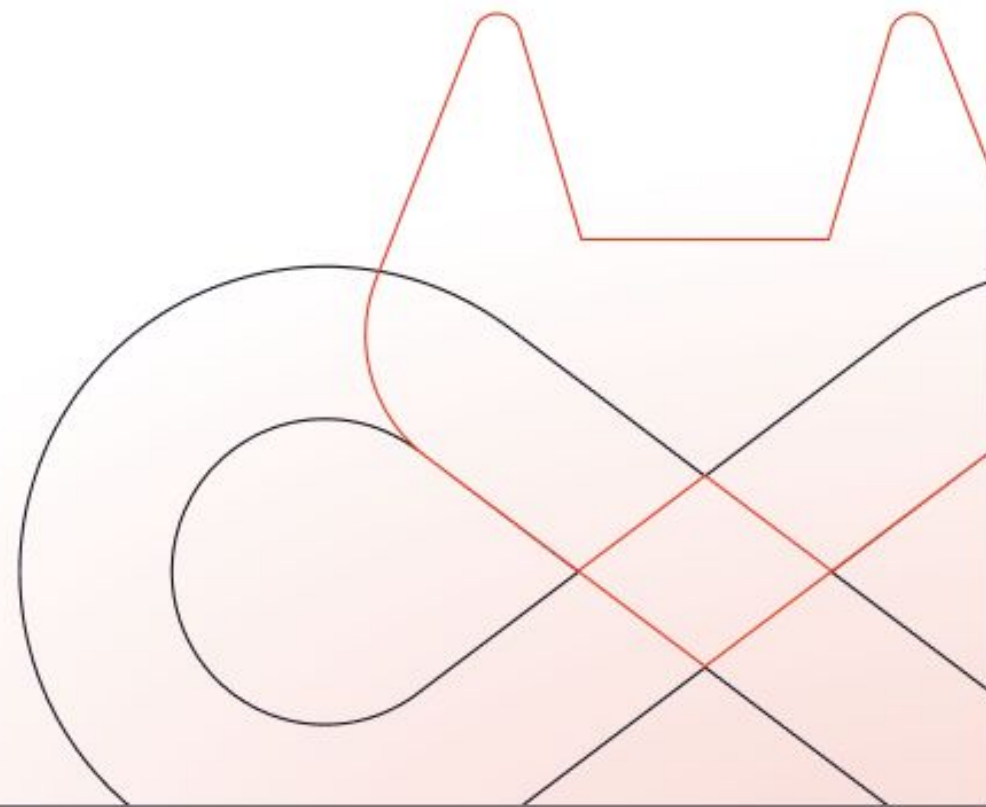


```
workflow:  
  rules:  
    - if: $CI_COMMIT_REF_NAME =~ /-wip$/  
      variables:  
        DOCKER_FILE: test.dockerfile  
    - if: $CI_COMMIT_REF_NAME != $CI_DEFAULT_BRANCH  
      variables:  
        DOCKER_FILE: main.dockerfile  
    - when: always
```

- The variable `DOCKER_FILE` will be set differently based on the rules.
- This pipeline will always run



# Artifacts





- Allows for **saving of build artifacts** and/or the output of any job
- Are available for use by **subsequent jobs**
- Can pull in **any combination of paths and/or files**
- Use **exclude** to **limit what is added**
- Use **depends** to **limit what gets downloaded** on subsequent jobs
- Use **when** to determine **if artifacts will be stored** or not
- Use **expire\_in** to determine **when artifacts will be destroyed**

```
artifacts:  
  when: on_success  
  paths:  
    - bin/target  
    - .exe  
  exclude:  
    - throw-away.exe  
  expire_in: 3mos
```



# Artifact download by URL



Three ways of  
accessing Artifacts  
via secure HTTPS

## **Build product for a specific CI pipeline run:**

```
https://gitlab.com/fpotter/examples/c-with-artifact/-/jobs/207917559/artifacts/file/helloworld
```

## **Build product for the latest CI pipeline run on a specific branch (main):**

```
https://gitlab.com/fpotter/examples/c-with-artifact/-/jobs/artifacts/main/raw/helloworld?job=build
```

## **All build products (as a Zip) for the latest CI pipeline run on a specific branch:**

```
https://gitlab.com/fpotter/examples/c-with-artifact/-/jobs/artifacts/main/download?job=build
```



# Artifact download in GitLab UI



## On the Pipelines page

Pipelines Jobs Environments Cycle Analytics

All 832 Running 0 Branches Tags Run pipeline CI Lint

| Status | Pipeline               | Commit                                      | Stages                  |                                                                                                                                                                         |
|--------|------------------------|---------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| passed | #6484362 by API latest | 57a5b7d5<br>Merge branch 'revert-caa47ca... | 00:13:35<br>5 hours ago |   |
| passed | #6484314 by API latest | 57a5b7d5<br>Merge branch 'revert-caa47ca... | 00:14:53<br>5 hours ago |   |

*Note: A red circle highlights the download icon in the first row, with a tooltip that says "Download compile artifacts".*

## On the Jobs page

Pipelines Jobs Environments Cycle Analytics

All 3,150 Pending 0 Running 0 Finished 2,989 CI Lint

| Status | Job                                  | Pipeline        | Stage  | Name    | Coverage                   |
|--------|--------------------------------------|-----------------|--------|---------|----------------------------|
| passed | #10500942 master<br>docker triggered | #6484362 by API | build  | compile | 05:04<br>about 5 hours ago |
| passed | #10500872 master<br>docker triggered | #6484314 by API | deploy | pages   | 05:08<br>about 5 hours ago |

*Note: A red circle highlights the download icon in the first row.*

## On a specific Job

Job artifacts

The artifacts will be removed in 6 days

Keep Download Browse

Job details

Duration: 5 minutes 4 seconds

Finished: about 5 hours ago

Runner: #44028

Raw Erase

*Note: A red circle highlights the "Download" button.*

## Artifact browser (for a Job)

Artifacts / selenium Download artifacts archive

| Name           | Size     |
|----------------|----------|
| ..             |          |
| selenium       |          |
| some-log.json  | 17 Bytes |
| some-page.html | 4 Bytes  |
| some-text.txt  | 4 Bytes  |



# Artifact administration



In a self-managed GitLab instance, job artifacts may be stored in **local storage** or **object storage**

Artifact **expiration times** can be configured at the instance level

Artifact downloads fall under GitLab's access control

## Project members permissions

**Note:** In GitLab 11.0, the Master role was renamed to Maintainer.

The following table depicts the various user permission levels in a project.

| Action                                                        | Guest | Reporter | Developer | Maintainer | Owner |
|---------------------------------------------------------------|-------|----------|-----------|------------|-------|
| Download project                                              | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| Leave comments                                                | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| View Insights charts <sup>?</sup>                             | ✓     | ✓        | ✓         | ✓          | ✓     |
| View approved/blacklisted licenses <sup>?</sup>               | ✓     | ✓        | ✓         | ✓          | ✓     |
| View license management reports <sup>?</sup>                  | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| View Security reports <sup>?</sup>                            | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| View project code                                             | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| Pull project code                                             | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| View GitLab Pages protected by <a href="#">access control</a> | ✓     | ✓        | ✓         | ✓          | ✓     |
| View wiki pages                                               | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| See a list of jobs                                            | ✓ (3) | ✓        | ✓         | ✓          | ✓     |
| See a job log                                                 | ✓ (3) | ✓        | ✓         | ✓          | ✓     |
| Download and browse job artifacts                             | ✓ (3) | ✓        | ✓         | ✓          | ✓     |
| Create new issue                                              | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| See related issues                                            | ✓     | ✓        | ✓         | ✓          | ✓     |
| Create confidential issue                                     | ✓ (1) | ✓        | ✓         | ✓          | ✓     |
| View confidential issues                                      | (2)   | ✓        | ✓         | ✓          | ✓     |
| Assign issues                                                 |       | ✓        | ✓         | ✓          | ✓     |



# Container and language-specific package registries



Docker registry

| Software repository                   | Description                                                                                                                          | Available in GitLab version |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <a href="#">Container Registry</a>    | The GitLab Container Registry enables every project in GitLab to have its own space to store <a href="#">Docker</a> images.          | 8.8+                        |
| <a href="#">Dependency Proxy</a> ⓘ    | The GitLab Dependency Proxy sets up a local proxy for frequently used upstream images/packages.                                      | 11.11+                      |
| <a href="#">Conan Repository</a> ⓘ    | The GitLab Conan Repository enables every project in GitLab to have its own space to store <a href="#">Conan</a> packages.           | 12.6+                       |
| <a href="#">Maven Repository</a> ⓘ    | The GitLab Maven Repository enables every project in GitLab to have its own space to store <a href="#">Maven</a> packages.           | 11.3+                       |
| <a href="#">NPM Registry</a> ⓘ        | The GitLab NPM Registry enables every project in GitLab to have its own space to store <a href="#">NPM</a> packages.                 | 11.7+                       |
| <a href="#">NuGet Repository</a> ⓘ    | The GitLab NuGet Repository will enable every project in GitLab to have its own space to store <a href="#">NuGet</a> packages.       | 12.8+                       |
| <a href="#">PyPi Repository</a> ⓘ     | The GitLab PyPi Repository will enable every project in GitLab to have its own space to store <a href="#">PyPi</a> packages.         | 12.10+                      |
| <a href="#">Go Proxy</a> ⓘ            | The Go proxy for GitLab enables every project in GitLab to be fetched with the <a href="#">Go proxy</a> protocol.                    | 13.1+                       |
| <a href="#">Composer Repository</a> ⓘ | The GitLab Composer Repository will enable every project in GitLab to have its own space to store <a href="#">Composer</a> packages. | 13.2+                       |

Dependency Proxy

Language-specific package registries





# dependencies



```
build:osx:
  stage: build
  script: make build:osx
  artifacts:
    paths:
      - binaries/

build:linux:
  stage: build
  script: make build:linux
  artifacts:
    paths:
      - binaries/

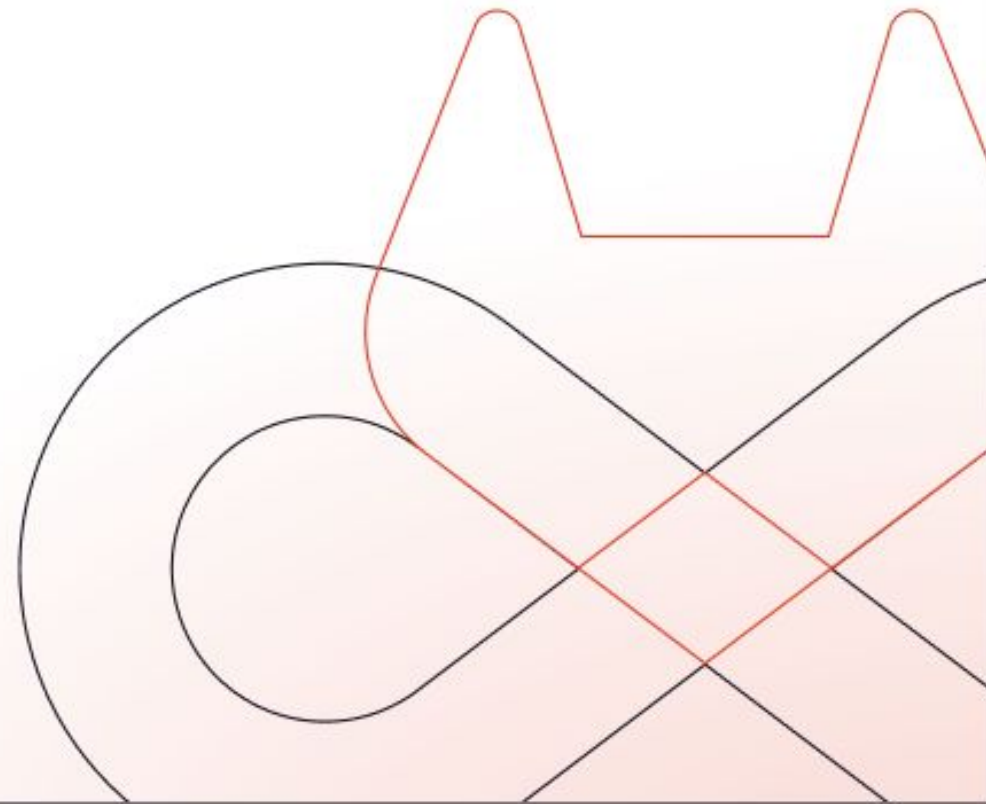
test:osx:
  stage: test
  script: make test:osx
  dependencies:
    - build:osx

test:linux:
  stage: test
  script: make test:linux
  dependencies:
    - build:linux

deploy:
  stage: deploy
  script: make deploy
```

- Reduce number of artifacts passed from previous jobs
- Improve job performance by preventing large artifacts to download for each job
- Empty array will skip downloading artifacts
- Only from jobs from previous stages

# Include and Extends



# Include



- Allows the inclusion of external YAML files
- Helps to break down the CI/CD configuration into multiple files and increases readability for long configuration files
- Possible to have template files stored in a central repository and projects include their configuration files
- Helps avoid duplicated configuration, for example, global default variables for all projects
- Can be nested (up to 100 includes)

```
include: '.gitlab-ci-production.yml'
```

```
include:  
  - local: '/templates/.gitlab-ci-template.yml'
```

```
include:  
  - project: 'my-group/my-project'  
    file: '/templates/.gitlab-ci-template.yml'
```

```
include:  
  - remote: 'https://gitlab.com/awesome-project/raw/master/.gitlab-ci-template.yml'
```

```
include:  
  - template: Android-Fastlane.gitlab-ci.yml  
  - template: Auto-DevOps.gitlab-ci.yml
```

| Method   | Description                                                    |
|----------|----------------------------------------------------------------|
| local    | Include a file from the local project repository.              |
| file     | Include a file from a different project repository.            |
| remote   | Include a file from a remote URL. Must be publicly accessible. |
| template | Include templates which are provided by GitLab.                |

# Extends



- Defines entry names that a job uses extends is going to inherit from
- Alternative to YAML anchors, and is more flexible and readable
- Performs a merge
- Supports multilevel inheritance (but more than 3 is not recommended)
- Can merge hidden jobs (.tests) or from regular jobs

```
.tests:
  script: rake test
  stage: test
  only:
    refs:
      - branches

rspec:
  extends: .tests
  script: rake rspec
  only:
    variables:
      - $RSPEC
```

Becomes

```
rspec:
  script: rake rspec
  stage: test
  only:
    refs:
      - branches
  variables:
    - $RSPEC
```



# Using Includes and Extends together



- Extends works across configuration files with include

If you have a local included.yml:

```
.template:  
  script:  
    - echo Hello!
```

You can do the following:

```
include: included.yml  
  
useTemplate:  
  image: alpine  
  extends: .template
```

This will run a job called `useTemplate` that runs `echo Hello!` as defined in the `.template` job, and uses the `alpine` Docker image as defined in the local job.



Q&A

